

# ASSIGNMENT-15.2

## Backend API Development: Creating RESTful services with AI

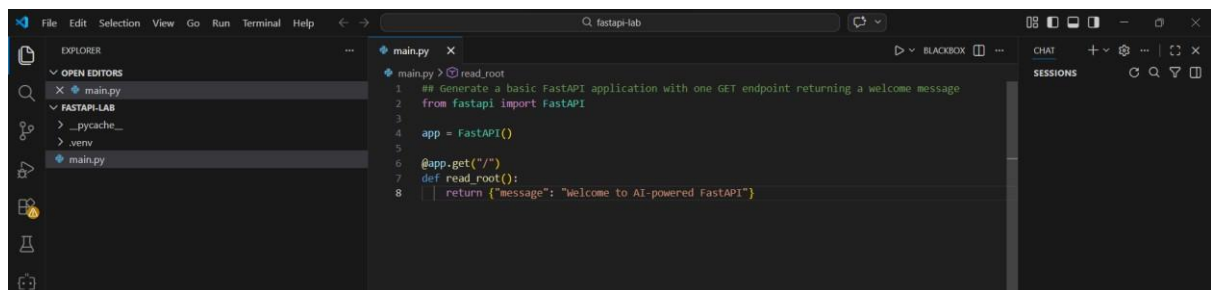
A.ABHIRAM

2403A51L13

BATCH:51

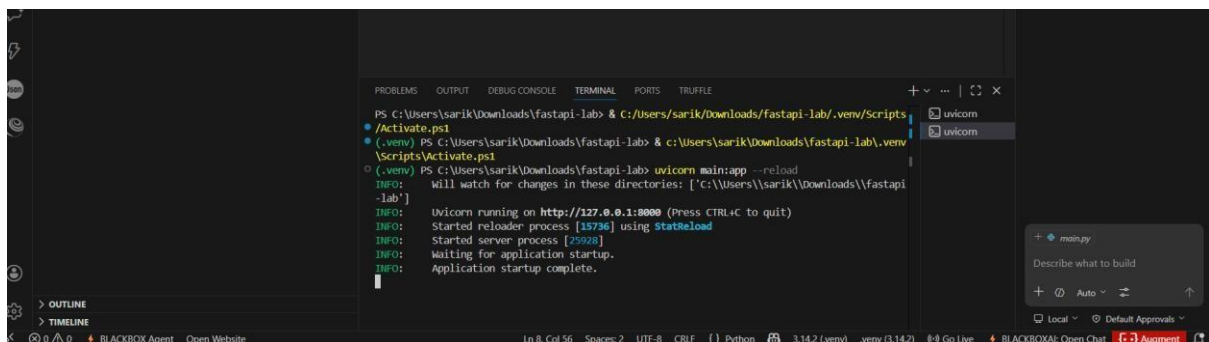
### Task Descriptor-1: (Setup Fast API Backend )

**PROMPT:** Generate a basic FastAPI application with one GET endpoint returning a welcome message **INPUT:**

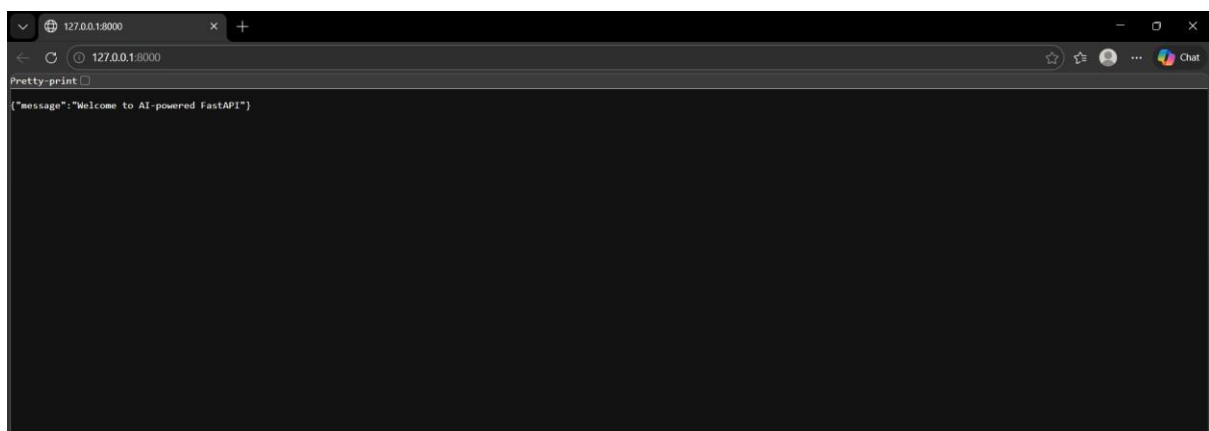


```
main.py > read_root
1 # Generate a basic FastAPI application with one GET endpoint returning a welcome message
2 from fastapi import FastAPI
3
4 app = FastAPI()
5
6 @app.get("/")
7 def read_root():
8     return {"message": "Welcome to AI-powered FastAPI"}
```

**OUTPUT:**



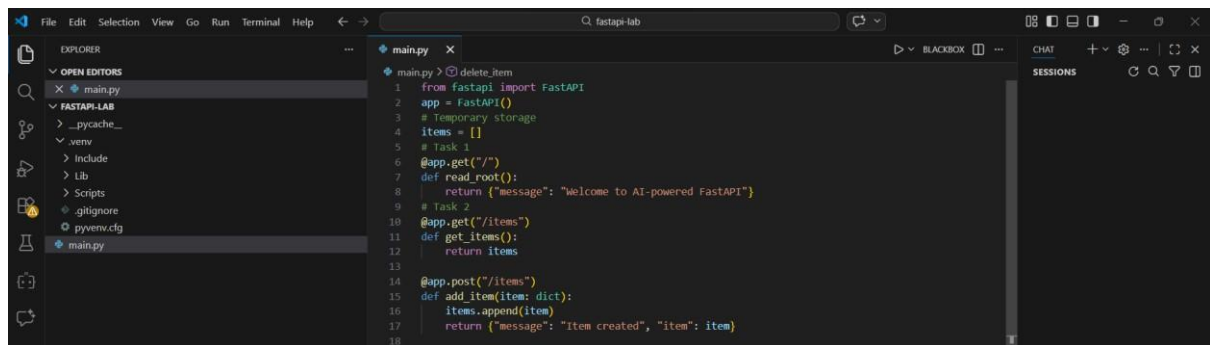
```
PS C:\Users\sarik\Downloads\fastapi-lab> & C:\Users\sarik\Downloads\fastapi-lab\.venv\Scripts\uvicorn
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> & C:\Users\sarik\Downloads\fastapi-lab\.venv\Scripts\uvicorn
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['c:\Users\sarik\Downloads\fastapi-lab']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reload process [15736] using Stateload
INFO: Started server process [25028]
INFO: Waiting for application startup.
INFO: Application startup complete.
```



```
{
  "message": "Welcome to AI-powered FastAPI"
}
```

**Explanation:** FastAPI uses decorators to define API routes easily. The root endpoint returns a JSON response when accessed. **Task Descriptor- 2: (API – Create and Read Items)**

**PROMPT:** Create FastAPI endpoints to add and retrieve items using a Python list **INPUT:**



```
1 from fastapi import FastAPI
2 app = FastAPI()
3 # Temporary storage
4 items = []
5 # Task 1
6 @app.get("/")
7 def read_root():
8     return {"message": "Welcome to AI-powered FastAPI"}
9 # Task 2
10 @app.get("/items")
11 def get_items():
12     return items
13
14 @app.post("/items")
15 def add_item(item: dict):
16     items.append(item)
17     return {"message": "Item created", "item": item}
```

**OUTPUT:**

```
10 @app.get("/items")
11 def get_items():
12     return items

PS C:\Users\sarik\Downloads\fastapi-lab & C:\Users\sarik\Downloads\fastapi-lab\.venv\Scripts\Activate.ps1
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> pip install fastapi uvicorn
Requirement already satisfied: fastapi in .\venv\Lib\site-packages (0.135.1)
Requirement already satisfied: uvicorn in .\venv\Lib\site-packages (0.42.0)
Requirement already satisfied: starlette>=0.46.0 in .\venv\Lib\site-packages (from fastapi) (0.52.1)
Requirement already satisfied: pydantic>=2.7.0 in .\venv\Lib\site-packages (from fastapi) (2.12.5)
Requirement already satisfied: typing-extensions>=4.8.0 in .\venv\Lib\site-packages (from fastapi) (4.15.0)
Requirement already satisfied: typing-inspection>=0.4.2 in .\venv\Lib\site-packages (from fastapi) (0.4.2)
Requirement already satisfied: annotated-doc>=0.0.2 in .\venv\Lib\site-packages (from fastapi) (0.0.4)
Requirement already satisfied: click>=7.0 in .\venv\Lib\site-packages (from uvicorn) (8.3.1)
Requirement already satisfied: h11>=0.8 in .\venv\Lib\site-packages (from uvicorn) (0.16.0)
Requirement already satisfied: colorama in .\venv\Lib\site-packages (from click>=7.0->uvicorn) (0.4.6)
Requirement already satisfied: annotated-types>=0.6.0 in .\venv\Lib\site-packages (from pydantic>=2.7.0->fastapi) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.5 in .\venv\Lib\site-packages (from pydantic>=2.7.0->fastapi) (2.41.5)
Requirement already satisfied: anyio<=3.6.2 in .\venv\Lib\site-packages (from starlette>=0.46.0->fastapi) (4.12.1)
Requirement already satisfied: idna>=2.8 in .\venv\Lib\site-packages (from anyio<=3.6.2->starlette>=0.46.0->fastapi) (3.11)
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\Users\sarik\Downloads\fastapi-lab']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [12616] using StatReload
```

```
Requirement already satisfied: annotated-types>=0.6.0 in .\venv\Lib\site-packages (from pydantic>=2.7.0->fastapi) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.5 in .\venv\Lib\site-packages (from pydantic>=2.7.0->fastapi) (2.41.5)
Requirement already satisfied: anyio<=3.6.2 in .\venv\Lib\site-packages (from starlette>=0.46.0->fastapi) (4.12.1)
Requirement already satisfied: idna>=2.8 in .\venv\Lib\site-packages (from anyio<=3.6.2->starlette>=0.46.0->fastapi) (3.11)
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\Users\sarik\Downloads\fastapi-lab']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [12616] using StatReload
INFO: Started server process [16640]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

POST

/items Add item

Cancel

Reset

Parameters

No parameters

Request body required

application/json

Edit Value Schema

```
{
  "name": "Pen",
  "price": 20
}
```

Execute

Responses

| Code | Description | Links |
|------|-------------|-------|
|------|-------------|-------|

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Pen",
    "price": 20
  }'
```

Request URL

http://127.0.0.1:8000/items

Server response

| Code | Details                                                                                                                                                                                                                                                                                    |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 200  | <div>Response body</div> <div><pre>{   "message": "Item created",   "item": {     "name": "Pen",     "price": 20   } }</pre></div> <div>Response headers</div> <div><pre>content-length: 59 content-type: application/json date: Tue, 17 Mar 2020 15:27:29 GMT server: uvicorn</pre></div> |

Responses

| Code | Description         | Links    |
|------|---------------------|----------|
| 200  | Successful Response | No links |

GET

/items Get items

Cancel

Parameters

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/items

Server response

| Code | Details                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 200  | <div>Response body</div> <div><pre>[   {     "name": "Pen",     "price": 20   },   {     "name": "Pen",     "price": 20   } ]</pre></div> |

Responses

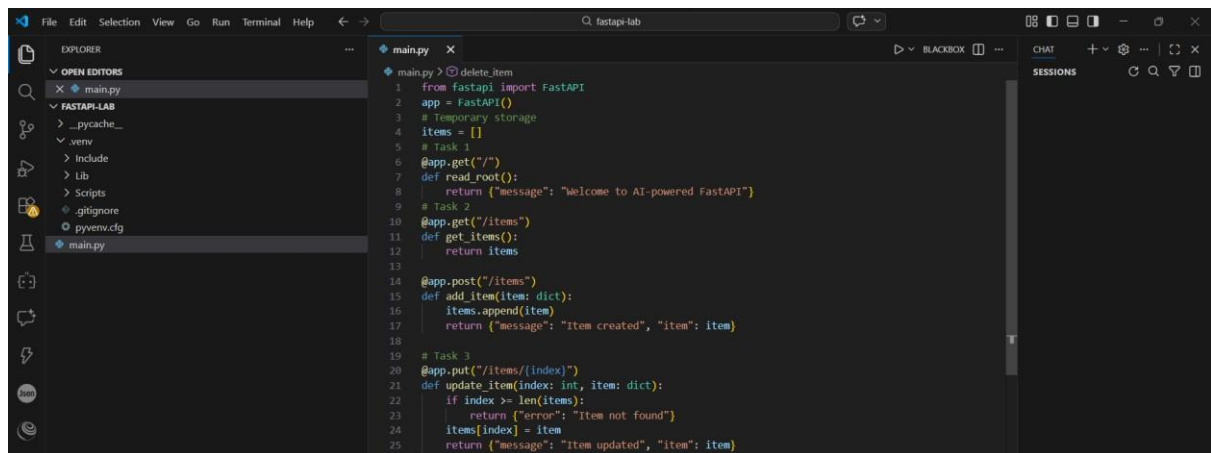
| Code | Description | Links |
|------|-------------|-------|
|------|-------------|-------|

**Explanation:** GET retrieves all items stored in the list. POST adds a new item into the list.

### Task Descriptor- 3: (Update Existing Item)

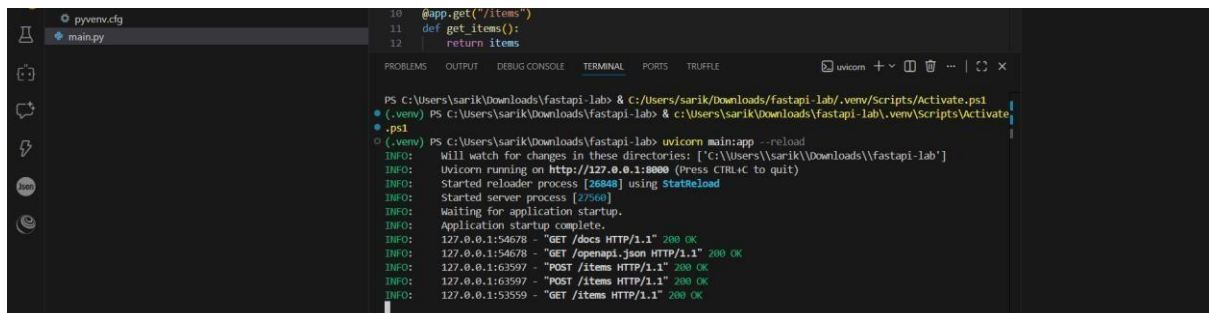
**PROMPT:** Create a PUT endpoint to update an item using its index.

**INPUT:**

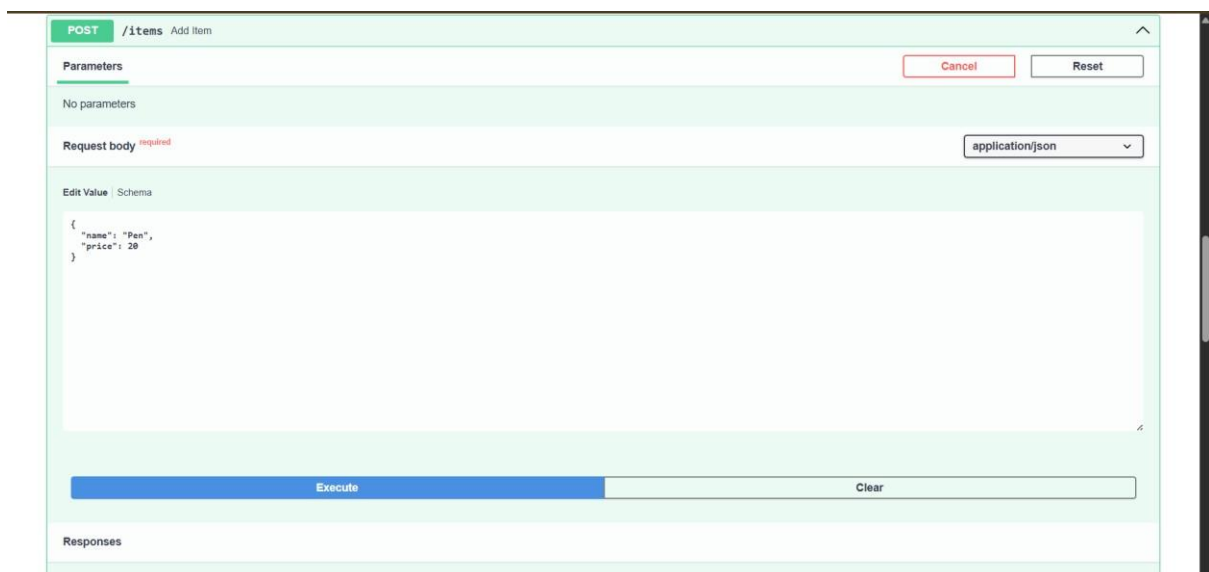


```
1 from fastapi import FastAPI
2 app = FastAPI()
3 # Temporary storage
4 items = []
5 # Task 1
6 @app.get("/")
7 def read_root():
8     return {"message": "Welcome to AI-powered FastAPI"}
9 # Task 2
10 @app.get("/items")
11 def get_items():
12     return items
13
14 @app.post("/items")
15 def add_item(item: dict):
16     items.append(item)
17     return {"message": "Item created", "item": item}
18
19 # Task 3
20 @app.put("/items/{index}")
21 def update_item(index: int, item: dict):
22     if index >= len(items):
23         return {"error": "Item not found"}
24     items[index] = item
25     return {"message": "Item updated", "item": item}
```

**OUTPUT:**



```
PS C:\Users\sarik\Downloads\fastapi-lab> & C:\Users\sarik\Downloads\fastapi-lab\.venv\Scripts\Activate.ps1
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> & C:\Users\sarik\Downloads\fastapi-lab\.venv\Scripts\Activate.ps1
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\Users\sarik\Downloads\fastapi-lab']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reload process [26848] using StarReload
INFO: Started server process [27560]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:54678 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:54678 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 127.0.0.1:163597 - "POST /items HTTP/1.1" 200 OK
INFO: 127.0.0.1:163597 - "POST /items HTTP/1.1" 200 OK
INFO: 127.0.0.1:153559 - "GET /items HTTP/1.1" 200 OK
```



**POST** /items Add item

Parameters Cancel Reset

No parameters

Request body required application/json

Edit Value Schema

```
{
  "name": "Pen",
  "price": 20
}
```

Execute Clear

Responses

GET /items Get Items

Parameters

Cancel

No parameters

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/items' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

CodeDetails

200

Response body

```
{
  "name": "Pen",
  "price": 10
},
{
  "name": "Pen",
  "price": 20
}
}
```

Download

PUT /items/{index} Update Item

Parameters

CancelReset

NameDescription

index \* required

integer (path)

0

Request body required

application/json

Edit ValueSchema

```
{
  "name": "Pencil",
  "price": 10
}
```

Execute

ExecuteClear

Responses

Curl

```
curl -X 'PUT' \
'http://127.0.0.1:8000/items/0' \
-H 'accept: application/json' \
-H 'Content-Type: application/json' \
-d '{
  "name": "Pencil",
  "price": 10
}'
```

Request URL

```
http://127.0.0.1:8000/items/0
```

Server response

CodeDetails

200

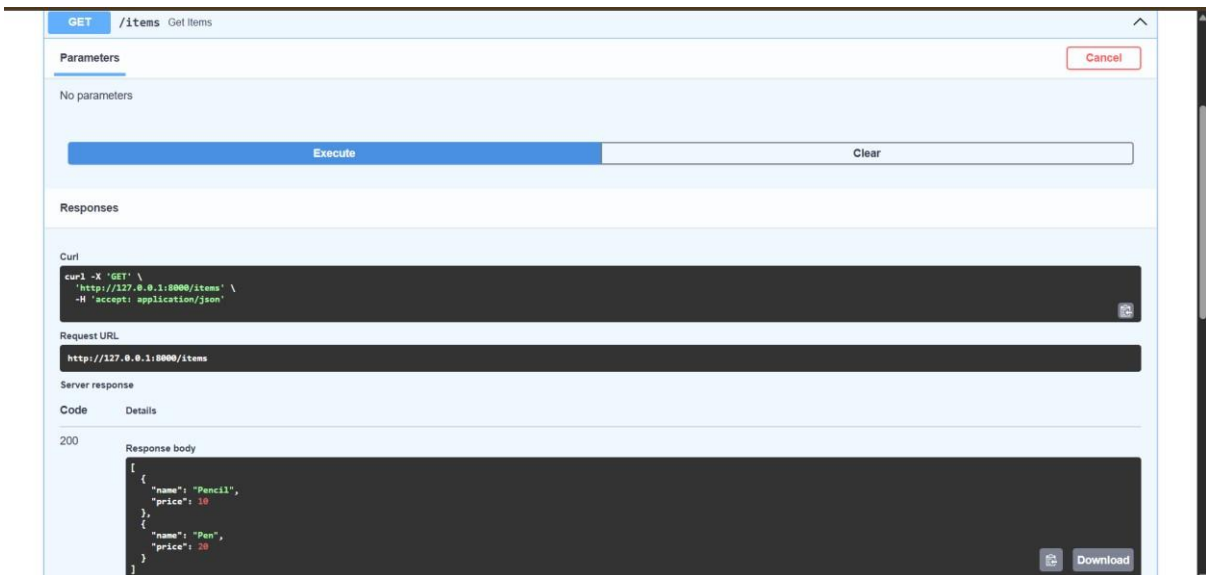
Response body

```
{
  "message": "Item updated",
  "item": {
    "name": "Pencil",
    "price": 10
  }
}
```

Download

Response headers

```
content-length: 62
content-type: application/json
date: Tue, 17 Mar 2026 15:56:10 GMT
server: uvicorn
```

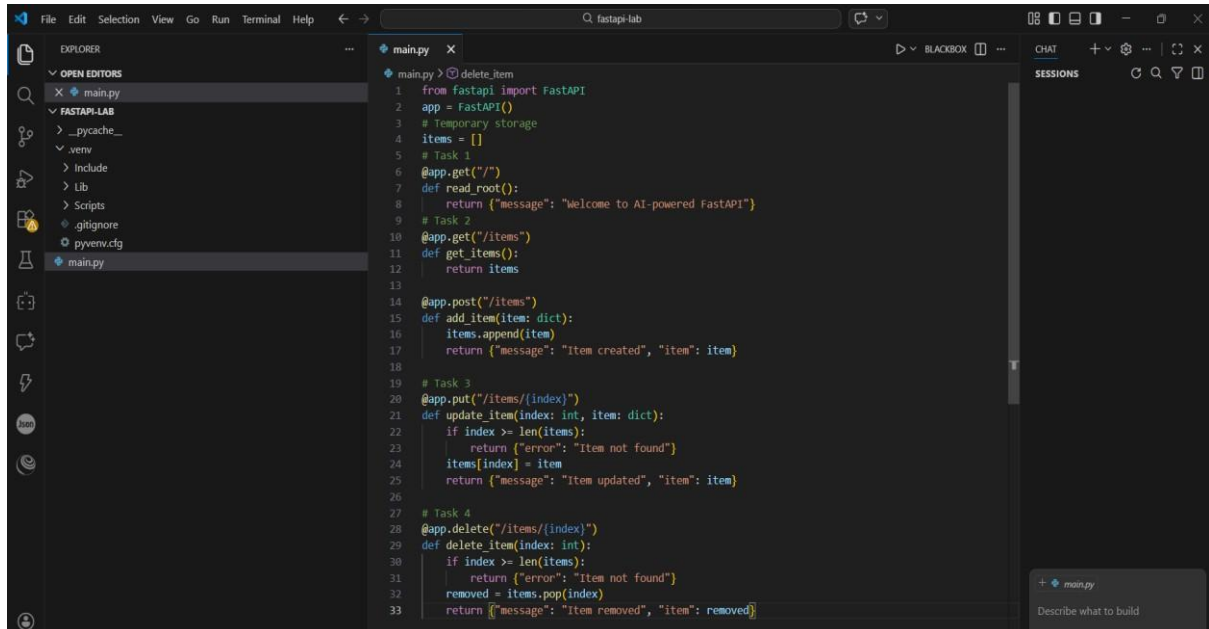


**Explanation:** PUT modifies an existing item in the list. It checks whether the given index exists.

#### Task Descriptor- 4: (Delete Item)

**PROMPT:** Create a DELETE endpoint to remove an item using its index.

**INPUT:**



**OUTPUT:**

```
PS C:\Users\sarik\Downloads\fastapi-lab> & C:\Users\sarik\Downloads\fastapi-lab\.venv\Scripts/Activate.ps1
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\Users\sarik\Downloads\fastapi-lab']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reload process [12756] using Sockshload
INFO: Started server process [21644]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [21644]
INFO: Stopping reload process [12756]
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab>
```

**POST** /items Add item

Parameters

No parameters

Request body *required* application/json

Edit Value | Schema

```
{
  "name": "Pen",
  "price": 20
}
```

Execute Clear

Responses

**GET** /items Get items

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

| Code | Details                                                                                                                            |
|------|------------------------------------------------------------------------------------------------------------------------------------|
| 200  | <p>Response body</p> <pre>{   "name": "Pencil",   "price": 10 }, {   "name": "Pen",   "price": 20 }, {   "name": "Ballpen" }</pre> |



DELETE

/items/{index} Delete Item

Cancel

Parameters

| Name                        | Description                    |
|-----------------------------|--------------------------------|
| index <span>required</span> |                                |
| Integer (path)              | <input type="text" value="0"/> |

ExecuteClear

Responses

Curl

curl -X 'DELETE' \n'http://127.0.0.1:8000/items/0' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:8000/items/0

Server response

| Code | Details                                                                                                                                                                                                                                                                                                                 |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 200  | <div><div>Response body</div><div>{\n  "message": "Item removed",\n  "item": {\n    "name": "Pencil",\n    "price": 10\n  }\n}</div><div>Download</div></div> <div><div>Response headers</div><div>content-length: 62\ncontent-type: application/json\ndate: Tue, 17 Mar 2026 16:01:00 GMT\nserver: uvicorn</div></div> |

Responses

| Code | Description         | Links    |
|------|---------------------|----------|
| 200  | Successful Response | No links |

Media type

application/json

Controls Accept header

GET

/items Get Items

Cancel

Parameters

No parameters

ExecuteClear

Responses

Curl

curl -X 'GET' \n'http://127.0.0.1:8000/items' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:8000/items

Server response

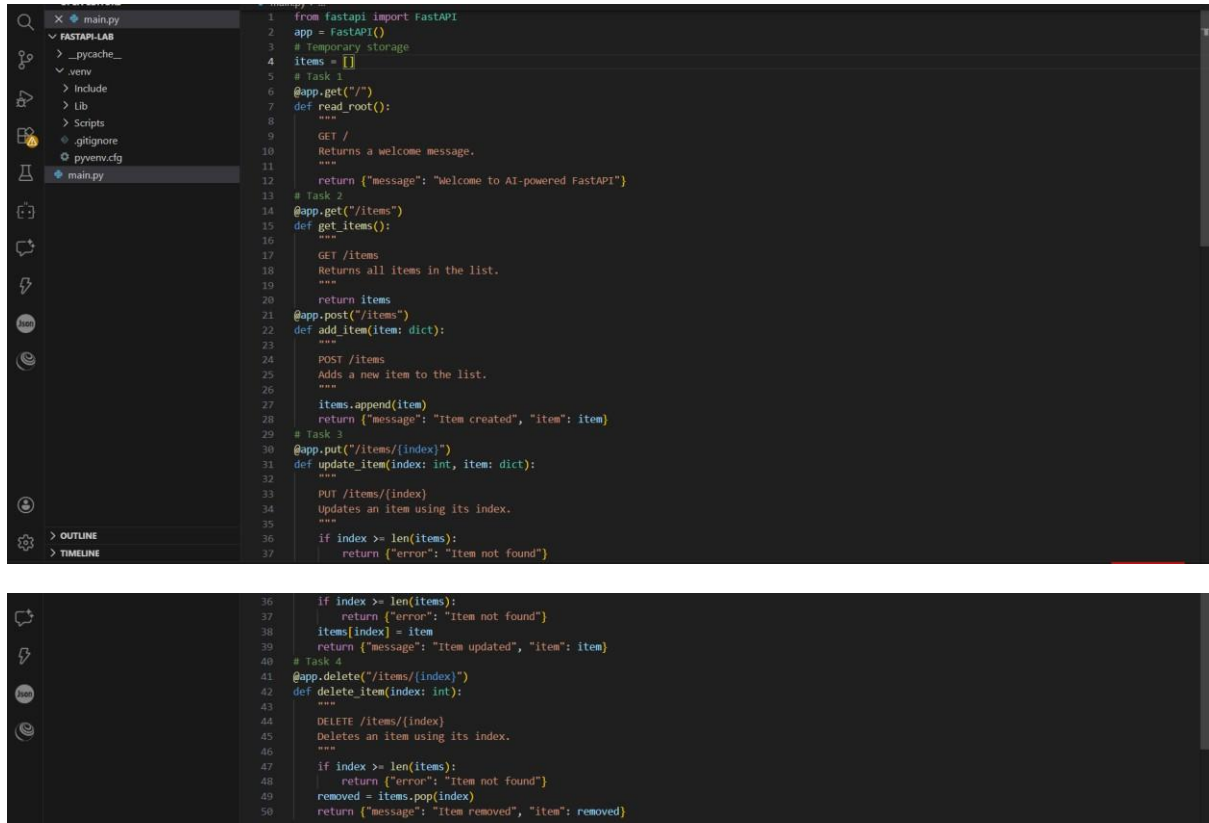
| Code | Details                                                                                                                                           |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 200  | <div><div>Response body</div><div>{\n  {\n    "name": "Pen",\n    "price": 20\n  },\n  {\n    "name": "Pen",\n    "price": 20\n  }\n}</div></div> |

**Explanation:** DELETE removes an item from the list. It returns the deleted item as response.

## Task Descriptor -5: (Add Auto-Generated Documentation)

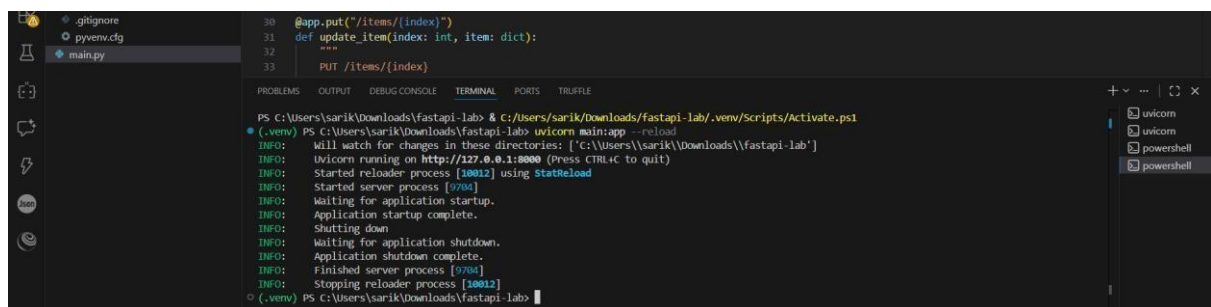
**PROMPT:** Add docstrings to endpoints for automatic Swagger documentation.

**INPUT:**



```
1 from fastapi import FastAPI
2 app = FastAPI()
3 # Temporary storage
4 items = []
5 # Task 1
6 @app.get("/")
7 def read_root():
8     """
9     GET /
10    Returns a welcome message.
11    """
12    return {"message": "Welcome to AI-powered FastAPI"}
13 # Task 2
14 @app.get("/items")
15 def get_items():
16     """
17     GET /items
18    Returns all items in the list.
19    """
20    return items
21 @app.post("/items")
22 def add_item(item: dict):
23     """
24     POST /items
25    Adds a new item to the list.
26    """
27    items.append(item)
28    return {"message": "Item created", "item": item}
29 # Task 3
30 @app.put("/items/{index}")
31 def update_item(index: int, item: dict):
32     """
33     PUT /items/{index}
34    Updates an item using its index.
35    """
36    if index >= len(items):
37        return {"error": "Item not found"}
38    items[index] = item
39    return {"message": "Item updated", "item": item}
40 # Task 4
41 @app.delete("/items/{index}")
42 def delete_item(index: int):
43     """
44     DELETE /items/{index}
45    Deletes an item using its index.
46    """
47    if index >= len(items):
48        return {"error": "Item not found"}
49    removed = items.pop(index)
50    return {"message": "Item removed", "item": removed}
```

**OUTPUT:**



```
PS C:\Users\sarik\Downloads\fastapi-lab> & C:\Users\sarik\Downloads\fastapi-lab\.venv\Scripts\Activate.ps1
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\Users\sarik\Downloads\fastapi-lab']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [10012] using StatReload
INFO: Started server process [9704]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [9704]
INFO: Stopping reloader process [10012]
(.venv) PS C:\Users\sarik\Downloads\fastapi-lab>
```

default

|        |                |             |   |
|--------|----------------|-------------|---|
| GET    | /              | Read Root   | ⌵ |
| GET    | /items         | Get Items   | ⌵ |
| POST   | /items         | Add Item    | ⌵ |
| PUT    | /items/{index} | Update Item | ⌵ |
| DELETE | /items/{index} | Delete Item | ⌵ |

Schemas

HTTPValidationError > Expand all object

ValidationError > Expand all object

default

GET / Read Root

Parameters Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/

Server response

| Code | Details                                                                                          |
|------|--------------------------------------------------------------------------------------------------|
| 200  | <p>Response body</p> <pre>{   "message": "Welcome to AI-powered FastAPI" }</pre> <p>Download</p> |

Response headers

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/

Server response

| Code | Details                                                                                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 200  | <p>Response body</p> <pre>{   "message": "Welcome to AI-powered FastAPI" }</pre> <p>Download</p> <p>Response headers</p> <pre>content-length: 43 content-type: application/json date: Tue, 17 Mar 2020 15:24:45 GMT server: uvicorn</pre> |

Responses

| Code | Description         | Links    |
|------|---------------------|----------|
| 200  | Successful Response | No links |

Media type

application/json

Controls Accept header

Example Value | Schema

```
"string"
```

GET /items Get Items

Parameters

Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/items' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

CodeDetails

200

Response body

[ ]

Download

Response headers

```
content-length: 2
content-type: application/json
date: Tue, 17 Mar 2026 15:25:59 GMT
```

POST /items Add Item

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "name": "Pen",
  "price": 20
}
```

Execute

Responses

CodeDescriptionLinks

GET /items Get Items

Parameters

Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/items' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

CodeDetails

200

Response body

```
[
  {
    "name": "Pen",
    "price": 20
  },
  {
    "name": "Pen",
    "price": 20
  }
]
```

Download

PUT /items/{index} Update Item

Parameters

Cancel

Reset

| Name                        | Description |
|-----------------------------|-------------|
| index <span>required</span> |             |
| integer                     |             |
| (path)                      |             |

Request body required

application/json

Edit Value

Schema

```
{
  "name": "Pencil",
  "price": 10
}
```

Execute

GET /items Get Items

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/items' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items
```

Server response

| Code | Details                                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 200  | <div><div>Response body</div><div><pre>{   "name": "Pencil",   "price": 10 }, {   "name": "Pen",   "price": 20 }</pre></div><div>Download</div></div> |

DELETE /items/{index} Delete Item

Parameters

Cancel

| Name                        | Description |
|-----------------------------|-------------|
| index <span>required</span> |             |
| integer                     |             |
| (path)                      |             |

Execute

Clear

Responses

Curl

```
curl -X 'DELETE' \
'http://127.0.0.1:8000/items/0' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/items/0
```

Server response

| Code | Details |
|------|---------|
|------|---------|

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/items/0' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/items/0

Server response

| Code | Details                                                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 200  | <p>Response body</p> <pre>{   "message": "Item removed",   "item": {     "name": "Pencil",     "price": 10   } }</pre> <p>Response headers</p> <pre>content-length: 62 content-type: application/json date: Tue, 17 Mar 2025 16:01:00 GMT server: uvicorn</pre> |

Responses

| Code | Description         | Links    |
|------|---------------------|----------|
| 200  | Successful Response | No links |

Media type

application/json

Controls Accept header

GET /items Get Items

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/items' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/items

Server response

| Code | Details                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------|
| 200  | <p>Response body</p> <pre>{   {     "name": "Pen",     "price": 20   },   {     "name": "Pen",     "price": 20   } }</pre> |

**Explanation:** FastAPI uses docstrings to generate API documentation. Swagger UI is available at /docs.